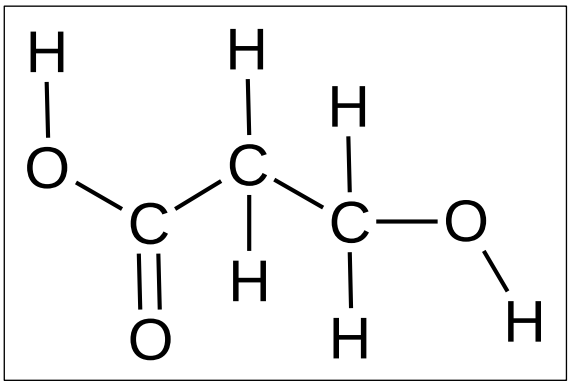
Answers Section A: Multiple Choice Questions

Answers to Organic Chemistry Past SAC questions:

Question 1

An organic compound is produced in the body during high levels of activity when the oxygen requirements of the muscles are greater than can be supplied by the blood circulation system. A representation of this compound is given in the diagram.



Which of the following reagents would you expect the above organic compound to **not react** with under the conditions indicated?

a) CH_3OH and H_2SO_4 catalyst

b) $H_2O_{(l)}$ at room temperature

c) $Br_{2(aq)}$ at room temperature

d) $MnO_4^-{}_{(aq)}$ and $H^+{}_{(aq)}$

Question 2

How many isomers are there for $C_3H_6Cl_2$?

a) 4

b) 5

c) 6

d) 3

Question 3

Two compounds which have the same molecular formula but different structural formulas are

a) 1-hexanol and pentanoic acid

b) Ethyl ethanoate and propanoic acid

c) Dichloroethane and 2-chlorobut-1-ene

d) 2-chlorobutane and 2-methyl-2-chloropropane

The following information is referred to in Questions 4 and 5 below.
2 molecules of HOOC – CH_2 – COOH combine with 1 molecule of HO – CH_2 – OH

Question 4

The best representation of the product is:

a) OOC – CH_2 – CH_2 – COO – CH_2 – OOC – CH_2 – CH_2 – COO

b) HOOC – CH_2 – COO – CH_2 – OOC – CH_2 – COOH

c) HOOC – CH_2 – CH_2 – COO – CH_2 – OOC – CH_2 – CH_2 – COOH

d) HO – CH_2 – CH_2 – COO – CH_2 – OOC – CH_2 – CH_2 – OH

Question 5

The best description of the other product(s) in this reaction is:

a) 1 molecule of water

b) 2 molecules of water

c) 3 molecules of water

d) 4 molecules of water

Question 6
A VCE student performs two tests on an organic compound. In the first test, a few drops of bromine solution were added to the compound. The colour of the solution remained red. In the second test, 0.5 mole of the compound was completely reacted with oxygen and 2 mole of carbon dioxide was produced. The formula of the compound is:
a) C_3H_8

b) C_4H_6

c) C_4H_{10}

d) C_4H_8

Question 7
For which one of the following molecular formulae is there only one possible structure?
a) $C_2H_2Cl_2$

b) $C_2H_4O_2$

c) C_2H_6O

d) C_2HCl_3

Question 8
To assist in working out the molecular formula of a compound with empirical formula CH_2O, its relative molecular mass was determined experimentally.
A 1.45 g sample of the compound was vaporised. The vapour occupied a volume of 484 mL at 1.25×10^5 Pa and 179°C.

The molecular formula of the compound was

a) CH_2O

b) $C_2H_4O_2$

c) $C_3H_6O_3$

d) $C_4H_8O_4$

Question 9
A compound has the molecular formula $C_8H_{10}Br_2$. If its structure **does not** contain any **rings**, how many **carbon to carbon double** bonds must be present?
a) 3

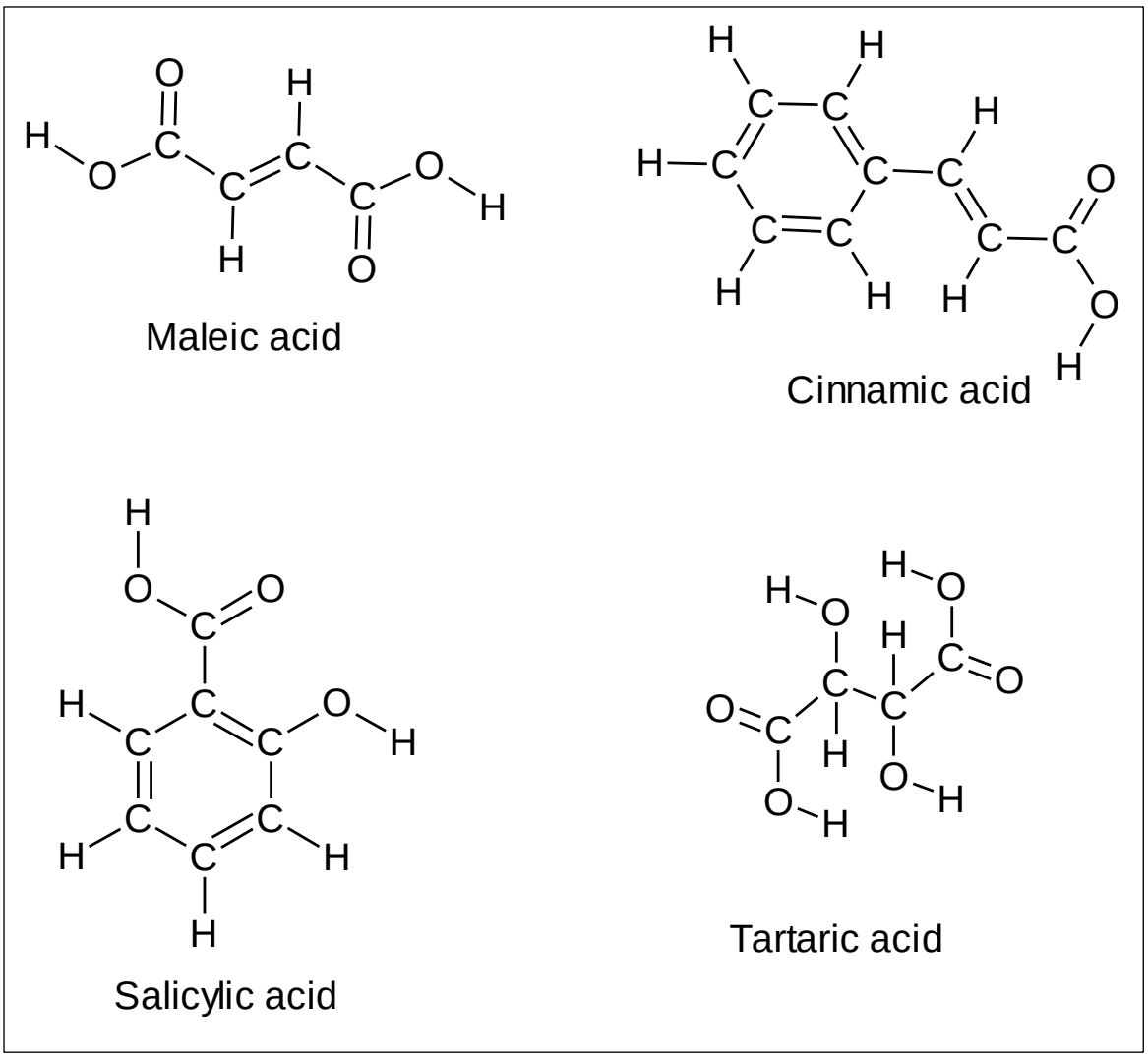
b) 4

c) 5

d) 2

Question 10

The structures of four different acids are shown below



Which of the following statements is correct?

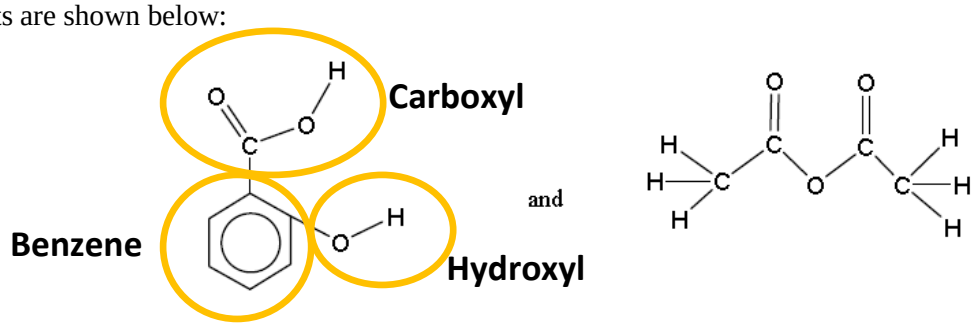
a) Cinnamic acid and maleic acid both could be involved in condensation polymerisation but not addition polymerisation.

b) Salicylic acid and cinnamic acid both contain the same number of functional groups.

c) 50.0 mL of 0.100 M solutions of maleic and cinnamic acid would require the same volume of 0.1 M NaOH for complete reaction.

d) Both Tartaric acid and maleic acid would react with Br_2 at room temperature.

Use the following information to answer Questions 11-13
Aspirin is produced by the reaction between salicylic acid and ethanoic (acetic) anhydride. The structures of these reactants are shown below:

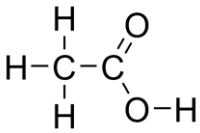


Question 11
Salicylic acid contains
A. a hydroxyl group, a cyclic hydrocarbon and an alkanol
B. two hydroxyl groups and a benzene ring
C. a carboxylic acid, a benzene ring and an alkanol
D. a carboxyl group, a hydroxyl group and a benzene ring

Answer is D

Question 12
When aspirin is produced by the reaction above, the second product is
A. ethanoic acid
B. acetic anhydride
C. methyl ethanoate
D. methanol

Answer is A



Question 13
Salicylic acid will react with a number of different molecules. Which of the following combination of reactant and reaction type is **not** possible with salicylic acid?
A. Ethanoic acid via condensation reaction
B. NaOH via acid-base reaction
C. Methanol via condensation reaction
D. Alanine via hydrolysis reaction

Answer is D: A hydrolysis reaction breaks links

Answers to Spectroscopy Past SAC questions:

Question 14
The most significant difference in the infrared spectra for the structural isomers, ethanoic acid and methyl methanoate, will be evident in the peaks due to
a) O-H bond stretching.

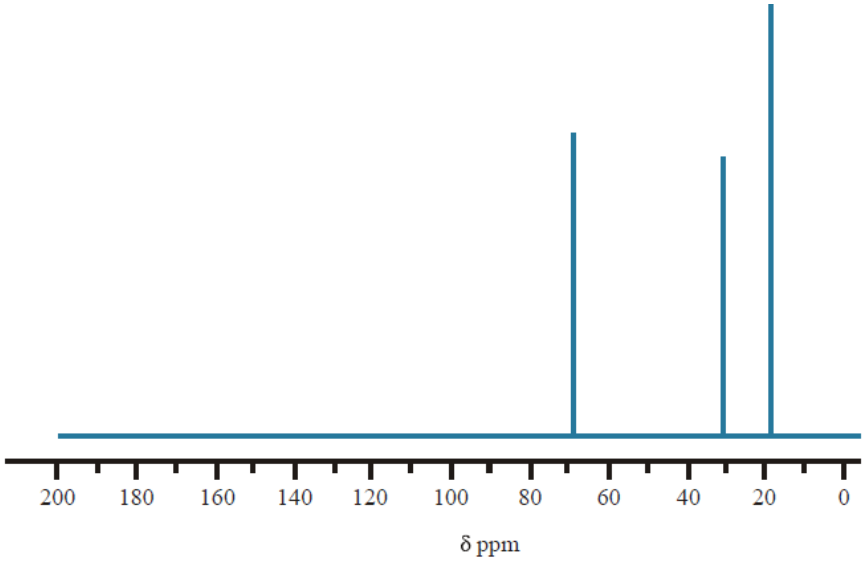
b) C=O bond stretching.

c) C-H bond stretching.

d) C-O bond stretching.

Question 15

Shown below is a representation of the ^{13}C NMR spectrum of a compound with molecular formula $C_4H_{10}O$.



This is the 13C NMR spectrum of

a) 2-propanol.

b) 2-butanol.

c) 2-methyl-propan-1-ol.

d) 2-methyl-propan-2-ol.

Question 16

Which of the spectral characteristics named below could **not** be used to distinguish between samples of **propanoic acid** and **methyl ethanoate**?

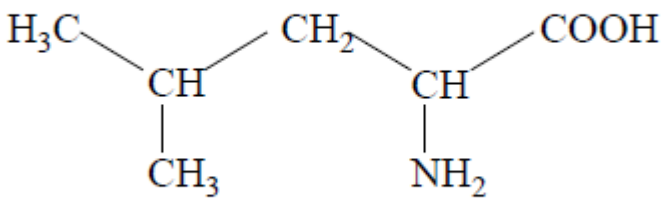
a) The number of peaks on the 1H NMR spectra

b) The region above wavenumber 1500 cm–1 on the infrared spectra

c) The major fragments detected on the mass spectra

d) The number of peaks on the 13C NMR spectra

Question 17
The structure of leucine is given in the diagram below**.**



leucine

The ^{13}C NMR and ^{1}H NMR of this compound will show the number of peaks corresponding to the number of **different carbon and hydrogen environments** respectively. These are:

a) 5 and 4

b) 5 and 6

c) 6 and 5

d) 6 and 6

Answers to HPLC Past SAC Questions: None available in 2013

Answers to Acid-Base and Redox Volumetric Analysis Past SAC Questions

Question 18

Barium Hydroxide is completely soluble in water. The pH at 25°C of a 0.0050 M solution of $Ba(OH)_2$ is

A. 2.0
B. 2.3
C. 11.7
D. 12.0

Question 19
In a separate experiment, excess aluminium is added to a 500 mL solution of HCl of pH 1. The volume of hydrogen gas formed at STP will be, in L,

A. 0.56

B. 1.12

C. 2.24

D. unable to be determined if the mass of aluminium is not given

Question 20
A piece of paper is found with the following data about a reaction between sodium carbonate, Na_2CO_3, and an unknown acid.

Mass of sodium carbonate used = 15.9 g
Volume of acid required = 150 mL
Concentration of acid = 2.00 M

Assuming the reaction goes to completion, it can be concluded from the data that

A. the acid is monoprotic
B. the acid is diprotic
C. this must be a weak acid
D. the sodium carbonate is very impure

Question 21
Sulfur dioxide (SO_2) is often added to wine as an antioxidant. A bottle of wine is labelled as containing 0.015% w/v of SO_2. The mass of SO_2 in a 750mL bottle of wine would be **closest** to

A. 110 mg
B. 50 mg
C. 5 mg
D. 200 mg

The following information refers to items 22 to 24

Phosphoric acid, H_3PO_4, normally contains trace quantities of arsenic. The amount of arsenic present is usually about 80 ppm (measured as As in H_3PO_4). Before the phosphoric acid can be used in the preparation of food additives the arsenic content must be reduced to less than 3 ppm.

Question 22
If 10 000 tonnes of phosphoric acid a year is converted into food additives, then each year the minimum amount of arsenic that must be removed is (1 tonne = 1000 kg)

A. 30 kg
B. 77 kg
C. 770 kg
D. 800 kg

Question 23
The maximum amount of arsenic that would still remain in 1 kg of the treated phosphoric acid would be

A. 3×10^{-6} g
B. 8×10^{-5} g
C. 3×10^{-3} g
D. 3.0 g

Question 24
If the arsenic is present in the original phosphoric acid as H_3AsO_3, then the mass of H_3AsO_3 in 1 tonne of the original acid would be approximately

A. 47 g
B. 77 g
C. 80 g
D. 134 g

Section B: Short AnswerQuestions

Organic Chemistry Past SAC questions

Question 1

Compound A is an ester composed of 43.3% oxygen by mass. When it undergoes complete combustion the mass of carbon dioxide produced is 22/9 times greater than the mass of water produced.

a. Determine the empirical formula of Compound A.

Assume 18 g of H_2O produced

$n(H_2O) = \frac{18}{18} = 1\,mol$

$n(CO_2) = \frac{18}{44} \times \frac{22}{9} = 1\,mol.$

C : H

$n(CO_2) : 2 \times n(H_2O)$

1 : 2

$\therefore CH_2$ (1M).

CH_2 : O

$\frac{56.7}{14} : \frac{43.3}{16}$

$\frac{4.05}{2.71} : \frac{2.71}{2.71}$

(1.5 : 1) × 2

3 : 2 (1M)

EF = $(CH_2)_3O_2$

$= C_3H_6O_2$ (1M).

3 mark

b. When 5.33 g of Compound A is vaporised, it occupies 2.22 L at 100°C and 1.00 x 10^5 Pa. Determine the molecular formula of compound A.

$PV = nRT$

$n = \frac{100\,kPa \times 2.22}{8.31 \times (273+100)}$

$= 0.0716\,mol$ (1M)

$M = \frac{5.33\,g}{0.0716\,mol}$

$= 74.4\,g/mol$ (1M)

$\frac{MF}{EF} = \frac{74.4}{74} = 1$

$\therefore MF = C_3H_6O_2$ (1M)

3 marks

Total = 6 marks

Question 2

a) Write the **systematic names** of the compounds, A, C, D and E in this chemical pathway.

A	Propene	[1 mark]
C	propan-1-ol	[1 mark]
D	2-methyl propene	[1 mark]
E	2-methyl propan-1-ol	[1 mark]

b) What reactants would have to be added for each of the following reactions, I to IV?

i)	NaOH or KOH	[1 mark]
ii)	$Cr_2O_7^{2-}{}_{(aq)}$ / $H^+{}_{(aq)}$ or $MnO4^-{}_{(aq)}$ / $H^+{}_{(aq)}$	[1 mark]
iii)	H_2O/H_3PO4	[1 mark]
iv)	$H_2SO_{4(l)}$ or concentrated H_2SO_4	[1 mark]

c) What type of chemical processes occurs in each of the following reactions

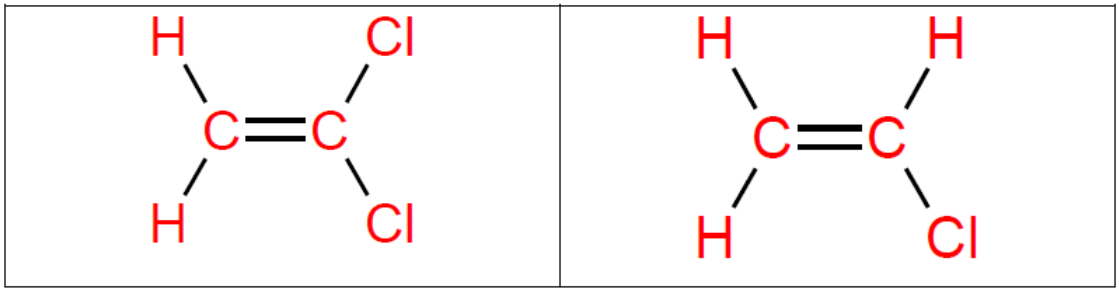
i)	Substitution reaction	[1 mark]
ii)	Oxidation reaction.	[1 mark]
iii)	Addition	[1 mark]
iv)	Condensation or Esterification reaction	[1 mark]

Question 3

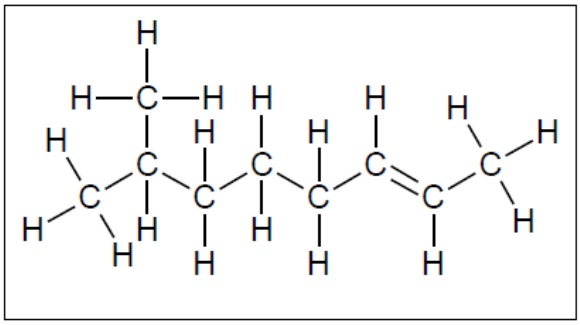
a) Saran is a copolymer made by polymerizing a mixture of two monomers. A section of Saran has the following structure:

$-CH_2 - CCl_2 - CH_2 - CHCl - CH_2 - CCl_2 - CH_2 - CHCl - CH_2 - CCl_2 - CH_2 - CHCl -$

Draw the structures of the monomers used to form this polymer. Show all bonds.

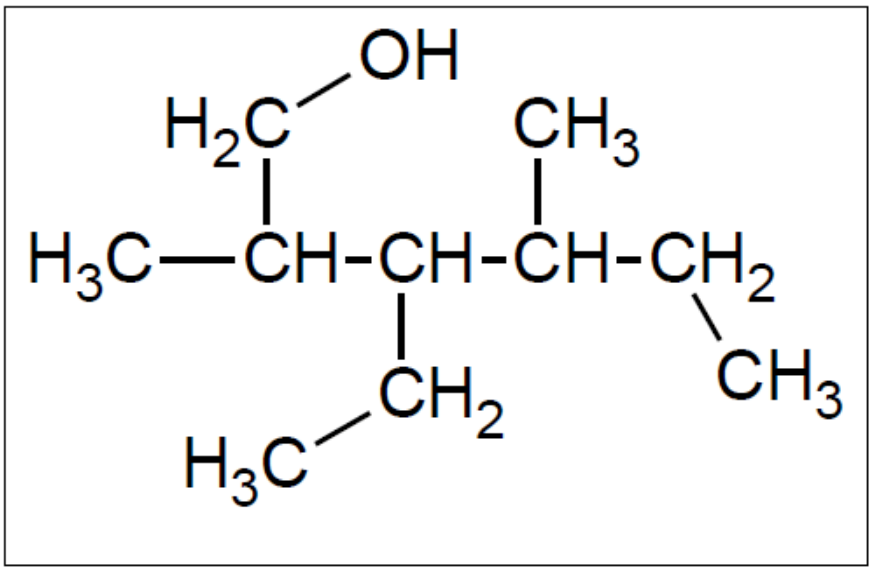


b) What is the systematic name for the hydrocarbon represented by the structure shown below?



7-methyloct-2-ene.

c) Draw the full structure of 2,4-dimethyl-3-ethyl-1-hexanol. Show all bonds



Question 4

a) Explain why propanoic acid has a higher boiling point than 1- propanol.

The stronger the intermolecular force the higher the boiling point. Alkanols are attracted to each other by dispersion forces and one hydrogen bond per molecule. Alkanoic acids are attracted to each other by dispersion force and two hydrogen bonds between molecules. Since alkanoic acids have a stronger intermolecular force they have a higher boiling point [1 mark]

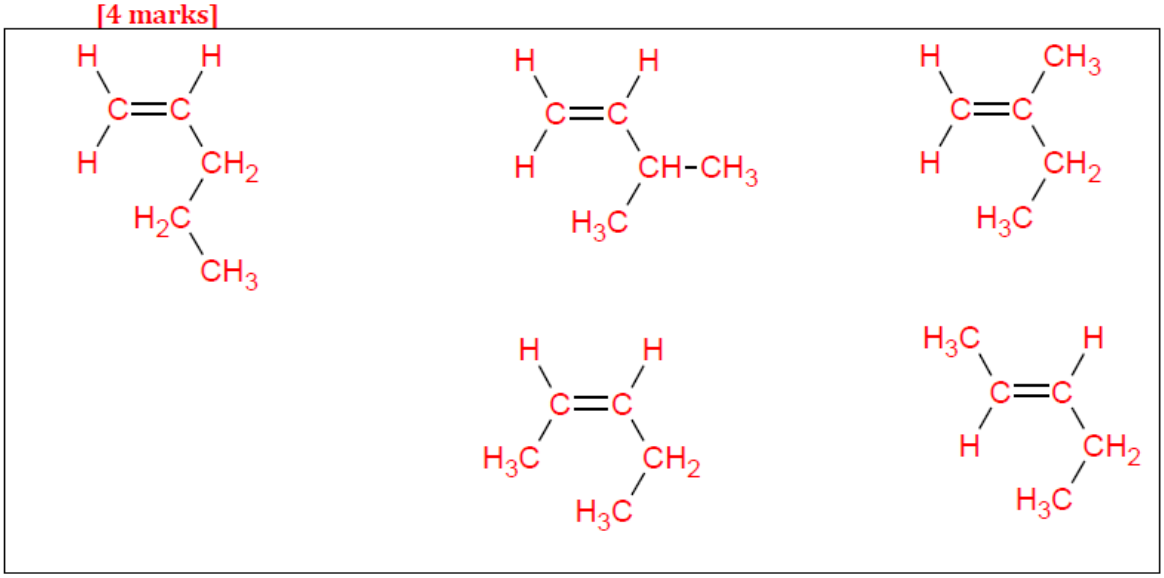
b) Explain why methyl methanoate is soluble in water whereas pentyl propanoate has a very low solubility in water.

Small esters such as methyl methanoate are soluble in water because some hydrogen bonding occurs with surrounding water molecules.

The hydrocarbon chain of bigger esters such as pentyl propanoate, are non-polar and only weak dispersion forces exists with surrounding water molecules. [1 mark]

c)

[4 marks]



d) (i) C_5H_{10} reacts readily with liquid iodochloride. Using the compound given in part (c), write a balanced equation (with states) for this reaction.

$$C_5H_{10\,(l)} + ICl_{(l)} \longrightarrow C_5H_{10}ICl_{(l)}$$

[1 mark for balanced equation and 1 mark for states]

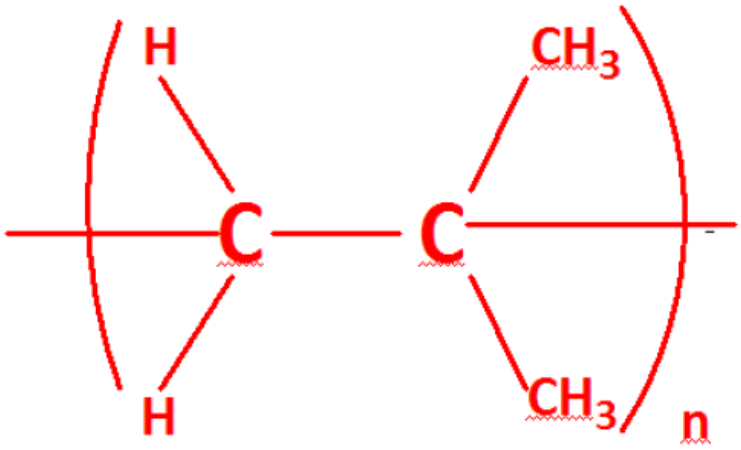
(ii) Write the systematic name of the product for this reaction.

2-chloro-2-methyl-4-iodobutane or 2-methyl-2-iodo-4-chlorobutane [1 mark]
**No consequential mark from part d(i)

e) What type of reaction is the reaction in (e)? **Addition reaction [1 mark]**

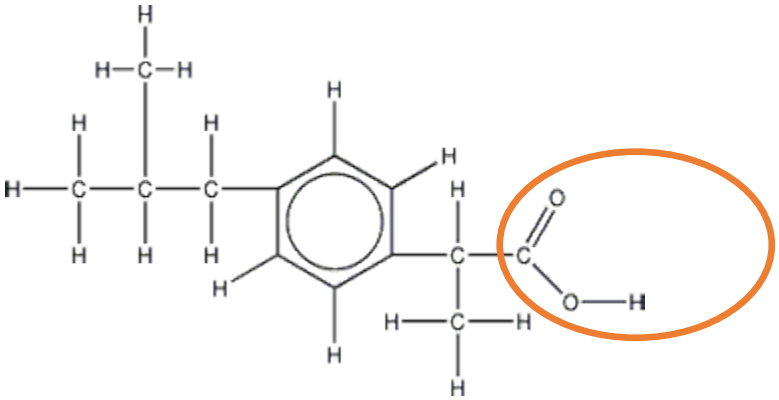
f)

2-methyl propene



Question 5

The structure of ibuprofen is shown below.

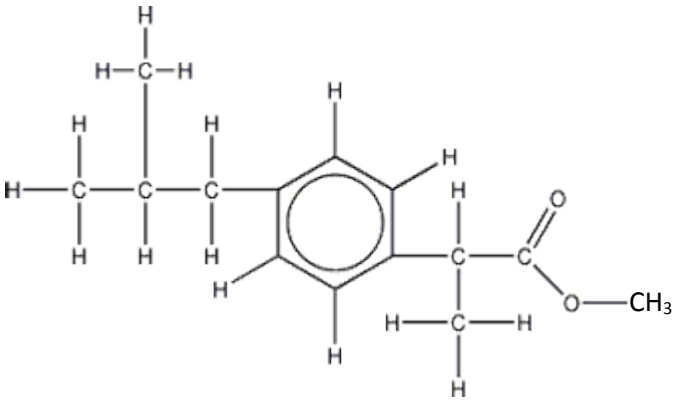


Carboxyl group (1 mark)

a. Identify one functional group in the structure above by **highlighting** or **circling** the functional group, and writing the **name** of the functional group.

1 mark

b. Draw the structure of the molecule that would result from a reaction between ibuprofen and methanol in the presence of concentrated sulphuric acid. (1 mark – accept semi-structural abbreviations)

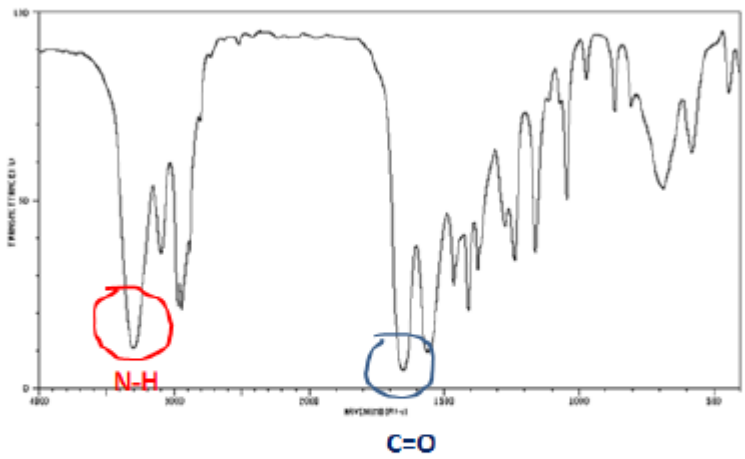


1 mark

Answers to Spectroscopy Past SAC questions:

Question 6

a) The infrared (IR) spectrum of this compound is given below:



Referring to the above spectrum, **circle and label** the absorption of the:

(i) N-H bond; and the

(ii) C=O bond.

[2 marks]

b)

(i) How many **different proton** environments are present? **4** **[1 mark]**

(ii) What is the **number** of protons in:

The peak areas indicate that the number of hydrogens for a:b:c:d is respectively 1:3:2:3 [1 mark]

(iii) A quartet is observed at 2.24 ppm. What information can you deduce from this splitting pattern?

A neighbouring C atom has 3 hydrogen atoms. Since these hydrogen atoms are in a different environment, a splitting pattern of (3 +1), a quartet is observed. [1 mark]

(iv) What is the singlet at 7.02 ppm due to?

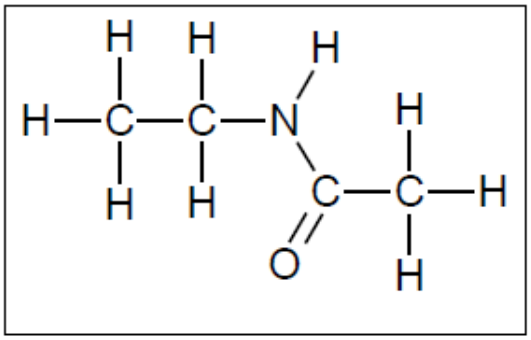
The singlet at 7.02 ppm indicates the presence of a –NH or –NHCO [1 mark]

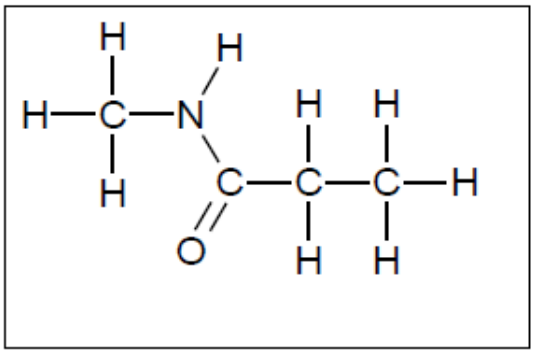
c)

A peak at 175 ppm is observed. What functional group or type of bond represents this peak?

There is a double covalent bond between a carbon atom and oxygen atom – single peak at 175 ppm [1 mark]

d) The compound has an amide functional (-NHCO-) group. Draw its structure, showing all bonds.





[1 mark] – Accept either molecule, as long as it is circled.

e) Write a balanced chemical equation for the formation of this compound. No states are required.

$CH_3NH_2 + CH_3CH_2COOH \longrightarrow CH_3NHCOCH_2CH_3 + H_2O$ **[1 mark]**

f) Classify the reaction in part (e)

Condensation, condensation elimination or acid-base reaction. [1 mark]

Question 7

a) (i) Identify the species that corresponds to the peaks at the following m/z values

m/z 200: $[^{79}BrCH_2CH_2CH_2{}^{79}Br]^+$
m/z 202: $[^{79}BrCH_2CH_2CH_2{}^{81}Br]^+$
m/z 204: $[^{81}BrCH_2CH_2CH_2{}^{81}Br]^+$
[1 mark for m/z 200 and m/z 202 ion; each ion is a singly positive ion; ignore the m/z 201 ion given in the question.

(ii) Write the equation (no states required) to show the formation of the ion at m/z 200.

$C_3H_6{}^{79}Br_2 + e \longrightarrow [C_3H_6{}^{79}Br_2]^+ + 2e$ **[1 mark]**
Pay consequential mark from (a)

b) Explain why the peaks at m/z 121 and 123 are similar in height (intensity)

79 Bromine and 81 Bromine have a relative abundance which is almost equal (approximately 50: 50) [1 mark]

Answers to HPLC Past SAC Questions: None available in 2013

Answers to Acid-Base and Redox Volumetric Analysis Past SAC Questions

Question 8

Zinc oxide is a major component of both sunscreens and anti-fungal creams. It can be manufactured from waste pieces of zinc in a process such as the one outlined below.

A 0.467 g piece of corroded zinc is dissolved in 50.0 mL of 2.00 M sulfuric acid. The following reaction occurs:

$$Zn\ (s) + H_2SO_4\ (aq) \longrightarrow ZnSO_4\ (aq) + H_2\ (g)$$

a. Which reactant is in excess? (show all workings)

$n(Zn) = \frac{0.467\,g}{65.4}$

$= 7.14 \times 10^{-3}$ mol (1M)

$n(H_2SO_4) = 2 \times \frac{50}{1000}$

$= 0.1$ mol (1M)

H_2SO_4 is in excess (1M)

ii. Under which circumstances would it be difficult to decide which chemical is in excess?

- when using an impure sample
- if reaction did not go to full completion
- if actual conc. of H_2SO_4 was less than specified amount (absorbed moisture).

2 + 1 = 3 marks

2 + 1 = 3 marks

A solution of sodium carbonate is now added to precipitate the zinc ions as zinc carbonate, $ZnCO_3$. The mass of precipitate formed is 0.223 g.

b. Write a balanced **ionic equation** for this reaction.

$Zn^{2+}\ (aq) + CO_3^{2-}\ (aq) \longrightarrow ZnCO_3\ (s)$

1 mark

The zinc carbonate is heated to decompose it to zinc oxide.

c. i. Write a balanced equation for this reaction.

$ZnCO_3\ (s) \longrightarrow ZnO\ (s) + CO_2\ (g)$ (1M) eqn (1M) states.

ii. Calculate the maximum possible mass of zinc oxide that could be obtained.

$n(ZnCO_3) = \frac{0.223\ g}{125.4}$

$= 1.778 \times 10^{-3}$ mol (1M)

$n(ZnO) = n(ZnCO_3)$

$m(ZnO) = 1.778 \times 10^{-3} \times 81.4$

$= 0.145\ g$ (1M)

2 + 2 = 4 marks

c. What was the % zinc in the original sample of corroded zinc you started with?

$m(Zn) = 1.778 \times 10^{-3}$ mol $\times 65.4$

$= 0.116\ g$ (1M)

$\%\ m(Zn) = \frac{0.116}{0.467} \times 100$

$= 24.9\%$ (1M)

2 marks

Total 10 marks

Question 9

A 40.00 mL sample of car battery acid containing sulfuric acid was diluted to 250.0 mL using distilled water.

20.00 mL aliquots were then pipetted into clean conical flasks and titrated against a 1.807 M solution of sodium hydroxide, the average titre for the titration was found to be 15.67 mL.

a. Write a balanced equation for the reaction above.

$H_2SO_4(aq) + 2NaOH(aq) \longrightarrow H_2O(l) + Na_2SO_4(aq)$

1 mark

b. Calculate the molar concentration of sulfuric acid in the **diluted** battery acid.

$n(NaOH) = C \times V = 1.807 \times 15.67 \times 10^{-3}$

$= 0.0283$ mol (1M)

$n(H_2SO_4) = \frac{1}{2} \times 0.0283 = 0.01416$ mol (1M)

$c(H_2SO_4) = \frac{0.01416}{20 \times 10^{-3}} = 0.7079\ M$ (1M)

3 marks

c. Calculate the molar concentration of sulphuric acid in the **original** battery acid sample.

$DF = \frac{250}{40}$ ∴ $C = \frac{250}{40} \times 0.7079 = \boxed{4.424\ M}$

1 mark

d. Several actions that could occur during this analytical procedure are listed below (**A–D**).

i. For each action, indicate the likely effect on the calculated concentration of sulfuric acid in the **original** battery acid by placing a tick in the appropriate box.

Action	Calculated result would be lower than the true value	No effect on calculated result	Calculated result would be higher than the true value
A. The volume of the **original sample** of battery acid was recorded incorrectly. The recorded volume was 0.15 mL less than the actual volume.			♥
B. The burette had been washed previously with distilled water only.			♥
C. The volumetric flask had been washed previously with distilled water but not dried.		♥	
D. A 15.00 mL pipette was unknowingly used instead of a 20.00 mL pipette.	♥		

ii. Explain your reasoning for the answer that you have given in the case of action **B**.

Conc. of NaOH would be lower than actual ∴ volume of titre would increase & calc. conc of H_2SO_4 would be higher than actual.

4 + 1 = 5 marks

Total = 10 marks